

Qualification Pack



Coal Bed Methane (CBM) Extraction Operator

Opencast/ Underground Coal

QP Code: MIN/Q1605

Version: 1.0

NSQF Level: 3.0

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MIN/Q1605: Coal Bed Methane (CBM) Extraction Operator

Brief Job Description

The individual in this role is responsible for operating and maintaining methane extraction unit to extract methane from the coal bed.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye-sight and not suffer from color-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N1613: Prepare methane extraction unit for operation](#)
2. [MIN/N1614: Perform extraction of methane from coal bed](#)
3. [MIN/N1615: Perform maintenance of the methane extraction unit](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Opencast

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 2: Underground Coal

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation

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Occupation	Specialized Operations
Country	India
NSQF Level	3.0
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3117.0200
Minimum Educational Qualification & Experience	10th grade pass OR 9th grade pass (with 1 year relevant experience) OR 8th grade pass (with 2 years relevant experience) OR 5th grade pass (with 5 years relevant experience) OR Previous relevant Qualification of NSQF Level (2.5-Jr. Coal Bed Methane Extraction Operator with 1.5 year of experience in relevant field)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	28/09/2026
NSQC Approval Date	29/09/2023
Version	1.0
Reference code on NQR	QG-03-MG-01026-2023-V1-SCMS
NQR Version	1.0

Remarks:

Total Duration: 420 Hours (360 hours Domain + 60 Hours Employability Skills)

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MIN/N1613: Prepare methane extraction unit for operation

Description

This unit is about inspecting & preparing the equipment for extracting methane from coal bed.

Scope

The scope covers the following :

- Prepare for work
- Inspect and prepare the methane extraction unit for operation

Elements and Performance Criteria

Prepare for work

To be competent, the user/individual on the job must be able to:

- PC1.** identify the work requirements from the instructions received from supervisor and interpreting information from previous day/shift logbook
- PC2.** use personal protective equipment appropriate for work activities
- PC3.** select appropriate type of auxiliary equipment for work activities

Inspect and prepare the methane extraction unit for operation

To be competent, the user/individual on the job must be able to:

- PC4.** check various controls, pressure gauges, pump and allied components, hydraulic fracture, compressor, gas separator and water separator, back pressure regulator, piping system etc. for proper functioning and any defect
- PC5.** ensure that all meters, gauges etc. are calibrated
- PC6.** check the direction of rotation of motor is corresponding to the regulated rotation direction of pump
- PC7.** check the condition of inlet and outlet valves and ensure that they are working properly and in normal state
- PC8.** check the pipe system and other parts such as anchor bolt, flange sealing gasket and other bolts for proper working and good condition
- PC9.** check all foundation bolts and outlet pipework bolts for appropriate tightness and condition
- PC10.** ensure all electrical, hydraulic, pneumatic, sealant and lubrication systems (as applicable) are connected and operational
- PC11.** maintain the operation and maintenance logbook to detail all activities conducted before starting the operation
- PC12.** report defects precisely to the supervisor if beyond scope of his/her role

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company

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- KU2.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** cost of equipment and loss for the company that results from damage of equipment
- KU5.** implications of delays in the process
- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to DTH/LH drilling operations in underground mines
- KU7.** different types of mines and detail of the mine one is working in
- KU8.** code of practice in specific areas of the mine
- KU9.** importance of safety in the vicinity of machinery
- KU10.** duties of workmen under the Mines Act-1952
- KU11.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU12.** international safety guidelines and policies like COSHH, OSHA etc.
- KU13.** the outcome of violation of safety procedures
- KU14.** functioning of methane extraction unit and its various components
- KU15.** normal running characteristics of methane extraction unit
- KU16.** how to inspect methane extraction unit and its components before operation
- KU17.** emergency response /disaster management plan prepared by the organization
- KU18.** signages, mining area-specific signs and other safety and emergence signals
- KU19.** basic computer work
- KU20.** how to use services such as NEFT, IMPS, UPI, RTGS for online banking
- KU21.** importance of entrepreneurship and characteristics of an effective entrepreneur
- KU22.** various campaigns run by government nationally such as Swachh Bharat Abhiyan, make in India campaign etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation

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GS13. report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for work</i>	6	12	-	3
PC1. identify the work requirements from the instructions received from supervisor and interpreting information from previous day/shift logbook	2	4	-	1
PC2. use personal protective equipment appropriate for work activities	2	4	-	1
PC3. select appropriate type of auxiliary equipment for work activities	2	4	-	1
<i>Inspect and prepare the methane extraction unit for operation</i>	24	38	-	17
PC4. check various controls, pressure gauges, pump and allied components, hydraulic fracture, compressor, gas separator and water separator, back pressure regulator, piping system etc. for proper functioning and any defect	3	5	-	1
PC5. ensure that all meters, gauges etc. are calibrated	2	4	-	2
PC6. check the direction of rotation of motor is corresponding to the regulated rotation direction of pump	3	5	-	2
PC7. check the condition of inlet and outlet valves and ensure that they are working properly and in normal state	3	4	-	2
PC8. check the pipe system and other parts such as anchor bolt, flange sealing gasket and other bolts for proper working and good condition	3	4	-	2
PC9. check all foundation bolts and outlet pipework bolts for appropriate tightness and condition	3	4	-	2
PC10. ensure all electrical, hydraulic, pneumatic, sealant and lubrication systems (as applicable) are connected and operational	2	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain the operation and maintenance logbook to detail all activities conducted before starting the operation	3	4	-	2
PC12. report defects precisely to the supervisor if beyond scope of his/her role	2	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1613
NOS Name	Prepare methane extraction unit for operation
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Specialized Operations
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQC Clearance Date	29/09/2023

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MIN/N1614: Perform extraction of methane from coal bed

Description

This unit is about performing processes for extracting methane gas from coal bed.

Scope

The scope covers the following :

- Start and operate the methane extraction unit
- Stop the methane extraction unit

Elements and Performance Criteria

Start and operate the methane extraction unit

To be competent, the user/individual on the job must be able to:

- PC1.** open the inlet valve and fully close the outlet valve before starting the pump
- PC2.** start the extraction unit (pump and separators) and slowly open the outlet valve until the pump reaches the desired flow
- PC3.** check the pressure gauge to ensure that the pump reaches the required outlet pressure
- PC4.** stop the pump, and take the corrective actions as per SOP
- PC5.** monitor the gauges for bearing temperature, excessive vibration, and noise during the operation
- PC6.** shut down the pump immediately, if there is no pressure or low pressure or pump exceeds normal levels and take the corrective action as per SOP
- PC7.** monitor the operation of gas separator and ensure that it is extracting the methane and water contaminants properly and transferring them through their respective pipes
- PC8.** monitor the process parameters by reading the various gauges and correct them if not within standards
- PC9.** check the oil level at regular intervals
- PC10.** observe any gas leakage during the operation and stop the extraction unit immediately, if found
- PC11.** monitor the operation of complete extraction unit and record the operational data as per the frequency in the control plan

Stop the methane extraction unit

To be competent, the user/individual on the job must be able to:

- PC12.** close the outlet valve, but ensure that the pump runs in this condition for no more than a few seconds
- PC13.** stop the complete extraction unit
- PC14.** maintain and complete all required documentation clearly, concisely as per SOP

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** relevant standards and procedures followed in the company
- KU2.** quality and damage checks to be done and importance of the same
- KU3.** risk and impact of not following defined procedures/work instructions
- KU4.** functioning of methane extraction unit and its various components
- KU5.** normal running characteristics of methane extraction unit
- KU6.** SOP recommended by the manufacturer for operating the extraction unit
- KU7.** how to start, operate and stop the extraction unit
- KU8.** how to maintain flow at inlet and outlet valves as per requirement
- KU9.** effect of improper flow, pressure, bearing temperature and oil levels on the extraction process
- KU10.** general safety rules during extraction unit operation
- KU11.** operation and safety aspects of the equipment
- KU12.** possible causes of common problems during extraction unit operation and remedies
- KU13.** signages, mining area-specific signs and other safety and emergence signals

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** read and interpret symbols and readings
- GS3.** write work related information in English/regional language
- GS4.** communicate the work requirements to the supervisor and co-workers
- GS5.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS6.** make decisions pertaining to the concerned area of work
- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Start and operate the methane extraction unit</i>	23	38	-	15
PC1. open the inlet valve and fully close the outlet valve before starting the pump	2	4	-	1
PC2. start the extraction unit (pump and separators) and slowly open the outlet valve until the pump reaches the desired flow	3	4	-	2
PC3. check the pressure gauge to ensure that the pump reaches the required outlet pressure	2	3	-	1
PC4. stop the pump, and take the corrective actions as per SOP	2	4	-	2
PC5. monitor the gauges for bearing temperature, excessive vibration, and noise during the operation	2	3	-	2
PC6. shut down the pump immediately, if there is no pressure or low pressure or pump exceeds normal levels and take the corrective action as per SOP	2	4	-	2
PC7. monitor the operation of gas separator and ensure that it is extracting the methane and water contaminants properly and transferring them through their respective pipes	2	3	-	1
PC8. monitor the process parameters by reading the various gauges and correct them if not within standards	2	3	-	1
PC9. check the oil level at regular intervals	2	4	-	1
PC10. observe any gas leakage during the operation and stop the extraction unit immediately, if found	2	3	-	1
PC11. monitor the operation of complete extraction unit and record the operational data as per the frequency in the control plan	2	3	-	1
<i>Stop the methane extraction unit</i>	7	12	-	5



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. close the outlet valve, but ensure that the pump runs in this condition for no more than a few seconds	2	4	-	2
PC13. stop the complete extraction unit	3	4	-	2
PC14. maintain and complete all required documentation clearly, concisely as per SOP	2	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1614
NOS Name	Perform extraction of methane from coal bed
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Specialized Operations
NSQF Level	3.0
Credits	5
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQC Clearance Date	29/09/2023

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MIN/N1615: Perform maintenance of the methane extraction unit

Description

This unit is about performing routine maintenance and troubleshooting tasks on the methane extraction unit.

Scope

The scope covers the following :

- Prepare for maintenance work
- Repair and maintain the methane extraction unit
- Perform post-maintenance activities

Elements and Performance Criteria

Prepare for maintenance work

To be competent, the user/individual on the job must be able to:

- PC1.** identify the extraction unit manufacturer's specifications from the user manual and drawings
- PC2.** read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance
- PC3.** identify and arrange the tools, consumables and spare parts required during the task
- PC4.** follow safety practices during maintenance activities as per organisational SOP

Repair and maintain the methane extraction unit

To be competent, the user/individual on the job must be able to:

- PC5.** perform basic health check-up of methane extraction unit as specified in the maintenance checklist
- PC6.** replenish coolants and lubricants as per SOP
- PC7.** clean dirt and dust from areas around close clearances, bearing housings, separators and motors
- PC8.** check coupling alignment and re-align if necessary
- PC9.** check the methane extraction unit parts i.e. motor, pump, gas separator, water separator, back pressure regulator, seals, auxiliary supplies e.g. heating/cooling (if fitted), wiring harness, connectors etc. for proper functioning to find out root cause of the problems like any leakage, short circuit in parts, breakage of wires and clamps, etc. and discuss the same with senior if required
- PC10.** check for unusual noise vibration and bearing temperatures in the extraction unit
- PC11.** check the separators, pump and piping, seal chamber and stuffing box etc. for leaks
- PC12.** check the instrument panel and control gauge to identify out of range readings to find the defect/cause of failure
- PC13.** replace/change the spare parts and consumables of the extraction unit as per the schedule
- PC14.** ensure proper lock out and tag out of methane extraction unit to avoid any untoward incident triggered due to unknowingly operation of machine / system under maintenance

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PC15. assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel

Perform post-maintenance activities

To be competent, the user/individual on the job must be able to:

PC16. conduct trial run of the extraction unit and verify that specified parameters are attained with no abnormalities

PC17. dispose off waste materials such as failed parts/aggregates, as per organisation's policies

PC18. fill the daily, weekly and monthly maintenance/defect sheets as per the format lay down by the organization

PC19. record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. relevant standards and procedures followed in the company

KU2. quality and damage checks to be done and importance of the same

KU3. risk and impact of not following defined procedures/work instructions

KU4. functioning of slurry pump unit and its various systems and components

KU5. normal running characteristics of methane extraction unit

KU6. how to read maintenance schedules and checklists recommended by the manufacturer

KU7. how to read layout or drawing and wiring diagrams to understand its structure

KU8. Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required

KU9. how to collect and store consumables, spare parts, tools, etc. as per organizational procedures

KU10. use of appropriate PPE, material handling equipment and tools for completing the maintenance tasks

KU11. corrective actions for common faults and failures in methane extraction unit

KU12. documentation required regarding repair, maintenance and service performed

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read work instructions, equipment manuals and process documents

GS2. read and interpret symbols and readings

GS3. write work related information in English/regional language

GS4. communicate the work requirements to the supervisor and co-workers

GS5. effectively communicate, listen and comprehend the information given by various sources about the site

GS6. make decisions pertaining to the concerned area of work



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- GS7.** plan and organize the work order and tasks
- GS8.** recognise a workplace problem and take suitable action
- GS9.** follow instructions and work on areas of improvement identified
- GS10.** complete the assigned tasks timely
- GS11.** use reasoning skills to identify and resolve basic problems
- GS12.** analyze and detect any potential problems which could arise during operation
- GS13.** report to the supervisor or deal with a colleague individually, depending on the type of concern

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for maintenance work</i>	7	12	-	4
PC1. identify the extraction unit manufacturer's specifications from the user manual and drawings	2	3	-	1
PC2. read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance	2	3	-	1
PC3. identify and arrange the tools, consumables and spare parts required during the task	2	3	-	1
PC4. follow safety practices during maintenance activities as per organisational SOP	1	3	-	1
<i>Repair and maintain the methane extraction unit</i>	19	30	-	12
PC5. perform basic health check-up of methane extraction unit as specified in the maintenance checklist	2	3	-	1
PC6. replenish coolants and lubricants as per SOP	2	3	-	1
PC7. clean dirt and dust from areas around close clearances, bearing housings, separators and motors	2	3	-	1
PC8. check coupling alignment and re-align if necessary	2	3	-	1
PC9. check the methane extraction unit parts i.e. motor, pump, gas separator, water separator, back pressure regulator, seals, auxiliary supplies e.g. heating/cooling (if fitted), wiring harness, connectors etc. for proper functioning to find out root cause of the problems like any leakage, short circuit in parts, breakage of wires and clamps, etc. and discuss the same with senior if required	2	3	-	2
PC10. check for unusual noise vibration and bearing temperatures in the extraction unit	2	3	-	1
PC11. check the separators, pump and piping, seal chamber and stuffing box etc. for leaks	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check the instrument panel and control gauge to identify out of range readings to find the defect/cause of failure	2	3	-	1
PC13. replace/change the spare parts and consumables of the extraction unit as per the schedule	1	2	-	1
PC14. ensure proper lock out and tag out of methane extraction unit to avoid any untoward incident triggered due to unknowingly operation of machine / system under maintenance	1	2	-	1
PC15. assess when the problem is beyond his competence and report the problem to suitably qualified/designated and competent personnel	1	2	-	1
<i>Perform post-maintenance activities</i>	4	8	-	4
PC16. conduct trial run of the extraction unit and verify that specified parameters are attained with no abnormalities	1	2	-	1
PC17. dispose off waste materials such as failed parts/aggregates, as per organisation's policies	1	2	-	1
PC18. fill the daily, weekly and monthly maintenance/defect sheets as per the format lay down by the organization	1	2	-	1
PC19. record all repairs carried out, time taken and unplanned tasks encountered during the maintenance activities	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1615
NOS Name	Perform maintenance of the methane extraction unit
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Specialized Operations
NSQF Level	3.0
Credits	3
Version	1.0
Last Reviewed Date	29/09/2023
Next Review Date	28/09/2026
NSQC Clearance Date	29/09/2023

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

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- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQ Clearance Date	25/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass %

Qualification Pack

aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1613.Prepare methane extraction unit for operation	30	50	-	20	100	15
MIN/N1614.Perform extraction of methane from coal bed	30	50	-	20	100	25
MIN/N1615.Perform maintenance of the methane extraction unit	30	50	-	20	100	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	75

Elective: 1 Opencast

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	0	20	100	25
Total	30	50	-	20	100	25



Qualification Pack

Elective: 2 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	0	20	100	25
Total	30	50	-	20	100	25

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.